Version 7.4.0

PVsyst - Simulation report

Grid-Connected System

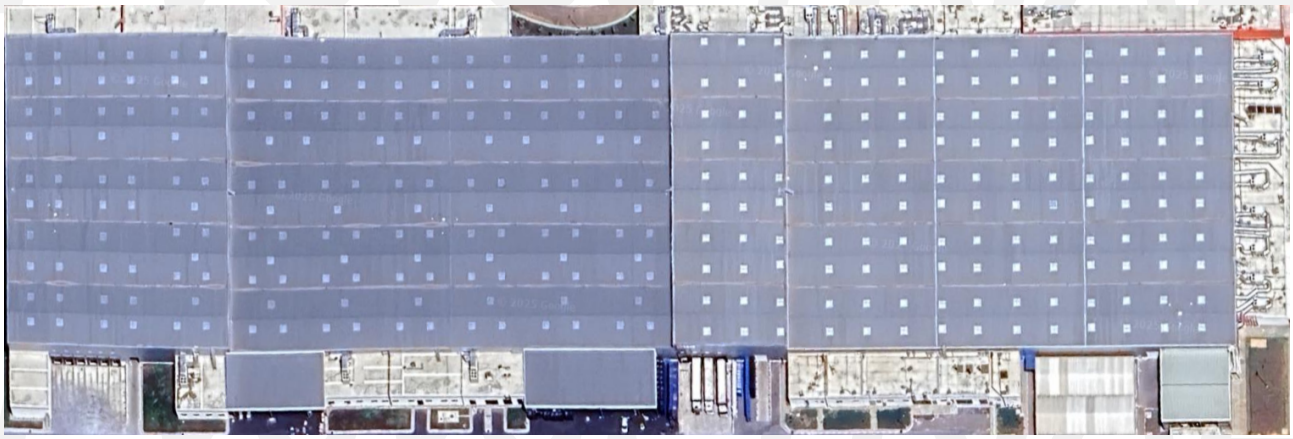
Projet: EST . OUEST

Variant: Nouvelle variante de simulation

No 3D scene defined, no shadings

System power: 1496 kWp

Haj Kaddour - Morocco



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Client

Entreprise

Zone Agropolis de Meknès

Meknès / 50080

Maroc

Signature

Author

Q384+RWP, Campus du savoir, Technopole med-est ,Commune Isly ,Oujda, O

Oujda / 60000

Maroc

0536705059

Signature

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Project summary

Geographical Site
Haj Kaddour
Morocco

Situation

Latitude	33.82 °N
Longitude	-5.43 °W
Altitude	643 m
Time zone	UTC

Project settings
Albedo 0.20

Meteo data
Haj Kaddour
Meteonorm 8.1 (1996-2015), Sat=100 % - Synthétique

System summary

Grid-Connected System
Simulation for year no 1

No 3D scene defined, no shadings

PV Field Orientation
Fixed planes 2 orientations
Tilts/azimuths 12 / -90 °
12 / 90 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	2580 units	Nb. of units	7 units
Pnom total	1496 kWp	Pnom total	1750 kWac
		Pnom ratio	0.855

Results summary

Produced Energy	1794401 kWh/year	Specific production	1199 kWh/kWp/year	Perf. Ratio PR	62.97 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		Sheds configuration	Models used	
Fixed planes	2 orientations	No 3D scene defined	Transposition	Perez
Tilts/azimuths	12 / -90 ° 12 / 90 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon	Near Shadings	User's needs
Free Horizon	No Shadings	Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM580M-7RL4-V
(Original PVsyst database)	
Unit Nom. Power	580 Wp
Number of PV modules	2580 units
Nominal (STC)	1496 kWp

Inverter

Manufacturer	Generic
Model	SG250-HX
(Original PVsyst database)	
Unit Nom. Power	250 kWac
Number of inverters	7 units
Total power	1750 kWac

Array #1 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #2 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #3 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

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PV Array Characteristics

Array #4 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #5 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #6 - Onduleur 1

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #7 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #8 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

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PV Array Characteristics

Array #9 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #10 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #11 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #12 - Onduleur 1

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	34 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	19.72 kWp	Total power	20.8 kWac
Modules	2 Strings x 17 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	17.99 kWp	Pnom ratio (DC:AC)	0.95
U mpp	683 V		
I mpp	26 A		

Array #13 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

PV Array Characteristics

Array #14 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

Array #15 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

Array #16 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

Array #17 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

Array #18 - Onduleur 2

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	32 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	18.56 kWp	Total power	20.8 kWac
Modules	2 Strings x 16 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	16.93 kWp	Pnom ratio (DC:AC)	0.89
U mpp	643 V		
I mpp	26 A		

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PV Array Characteristics

Array #19 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #20 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #21 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #22 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #23 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

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PV Array Characteristics

Array #24 - Onduleur 2

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #25 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #26 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #27 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #28 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

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PV Array Characteristics

Array #29 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #30 - Onduleur 3

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #31 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #32 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #33 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

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PV Array Characteristics

Array #34 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #35 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #36 - Onduleur 3

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #37 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #38 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

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PV Array Characteristics

Array #39 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #40 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #41 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #42 - Onduleur 4

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #43 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

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VC3, Simulation date:
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with v7.4.0

PV Array Characteristics

Array #44 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #45 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #46 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #47 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #48 - Onduleur 4

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

PVsyst V7.4.0
VC3, Simulation date:
26/11/25 23:26
with v7.4.0

PV Array Characteristics

Array #49 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #50 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #51 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #52 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #53 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

PVsyst V7.4.0
VC3, Simulation date:
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with v7.4.0

PV Array Characteristics

Array #54 - Onduleur 5

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #55 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #56 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #57 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #58 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

PVsyst V7.4.0
VC3, Simulation date:
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with v7.4.0

PV Array Characteristics

Array #59 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #60 - Onduleur 5

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #61 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #62 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #63 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

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PV Array Characteristics

Array #64 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #65 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #66 - Onduleur 6

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #67 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #68 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

PVsyst V7.4.0
VC3, Simulation date:
26/11/25 23:26
with v7.4.0

PV Array Characteristics

Array #69 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #70 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #71 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #72 - Onduleur 6

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #73 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst V7.4.0
VC3, Simulation date:
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with v7.4.0

PV Array Characteristics

Array #74 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #75 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #76 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #77 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #78 - Onduleur 7

Orientation	#2		
Tilt/Azimuth	12/90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

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VC3, Simulation date:
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PV Array Characteristics

Array #79 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #80 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #81 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #82 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Array #83 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

PVsyst TRIAL

PV Array Characteristics

Array #84 - Onduleur 7

Orientation	#1		
Tilt/Azimuth	12/-90 °		
Number of PV modules	30 units	Number of inverters	1 * MPPT 8% 0.1 unit
Nominal (STC)	17.40 kWp	Total power	20.8 kWac
Modules	2 Strings x 15 In series		
At operating cond. (50°C)		Operating voltage	500-1450 V
Pmpp	15.87 kWp	Pnom ratio (DC:AC)	0.84
U mpp	603 V		
I mpp	26 A		

Total PV power

Nominal (STC)	1496 kWp
Total	2580 modules
Module area	7054 m²

Total inverter power

Total power	1750 kWac
Nb. of inverters	7 units
	0.0 unused
Pnom ratio	0.86
No power sharing	

Array losses

Array Soiling Losses
Loss Fraction 4.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

LID - Light Induced Degradation
Loss Fraction 1.5 %

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.4 %

Module average degradation
Year no 1
Loss factor 0.4 %/year
Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor
ASHRAE Param.: IAM = 1 - bo (1/cosi -1)
bo Param. 0.40

Spectral correction
FirstSolar model
Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0,85914	-0,02088	-0,0058853	0,12029	0,026814	-0,001781

DC wiring losses

Global wiring resistance 4.6 mΩ
Loss Fraction 1.5 % at STC

Array #1 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

Array #2 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

Array #3 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

Array #4 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

Array #5 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

Array #6 - Onduleur 1
Global array res. 429 mΩ
Loss Fraction 1.5 % at STC

DC wiring losses

Array #7 - Onduleur 1		**Array #8 - Onduleur 1**	
Global array res.	429 mΩ	Global array res.	429 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #9 - Onduleur 1		**Array #10 - Onduleur 1**	
Global array res.	429 mΩ	Global array res.	429 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #11 - Onduleur 1		**Array #12 - Onduleur 1**	
Global array res.	429 mΩ	Global array res.	429 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #13 - Onduleur 2		**Array #14 - Onduleur 2**	
Global array res.	404 mΩ	Global array res.	404 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #15 - Onduleur 2		**Array #16 - Onduleur 2**	
Global array res.	404 mΩ	Global array res.	404 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #17 - Onduleur 2		**Array #18 - Onduleur 2**	
Global array res.	404 mΩ	Global array res.	404 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #19 - Onduleur 2		**Array #20 - Onduleur 2**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #21 - Onduleur 2		**Array #22 - Onduleur 2**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #23 - Onduleur 2		**Array #24 - Onduleur 2**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #25 - Onduleur 3		**Array #26 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #27 - Onduleur 3		**Array #28 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #29 - Onduleur 3		**Array #30 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #31 - Onduleur 3		**Array #32 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #33 - Onduleur 3		**Array #34 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #35 - Onduleur 3		**Array #36 - Onduleur 3**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #37 - Onduleur 4		**Array #38 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #39 - Onduleur 4		**Array #40 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #41 - Onduleur 4		**Array #42 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC

DC wiring losses

Array #43 - Onduleur 4		**Array #44 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #45 - Onduleur 4		**Array #46 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #47 - Onduleur 4		**Array #48 - Onduleur 4**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #49 - Onduleur 5		**Array #50 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #51 - Onduleur 5		**Array #52 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #53 - Onduleur 5		**Array #54 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #55 - Onduleur 5		**Array #56 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #57 - Onduleur 5		**Array #58 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #59 - Onduleur 5		**Array #60 - Onduleur 5**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #61 - Onduleur 6		**Array #62 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #63 - Onduleur 6		**Array #64 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #65 - Onduleur 6		**Array #66 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #67 - Onduleur 6		**Array #68 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #69 - Onduleur 6		**Array #70 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #71 - Onduleur 6		**Array #72 - Onduleur 6**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #73 - Onduleur 7		**Array #74 - Onduleur 7**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #75 - Onduleur 7		**Array #76 - Onduleur 7**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC
Array #77 - Onduleur 7		**Array #78 - Onduleur 7**	
Global array res.	378 mΩ	Global array res.	378 mΩ
Loss Fraction	1.5 % at STC	Loss Fraction	1.5 % at STC

DC wiring losses

Array #79 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

Array #80 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

Array #81 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

Array #82 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

Array #83 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

Array #84 - Onduleur 7
Global array res. 378 mΩ
Loss Fraction 1.5 % at STC

System losses

Unavailability of the system
Time fraction 2.0 %
7.3 days,
3 periods

PVsyst TRIAL

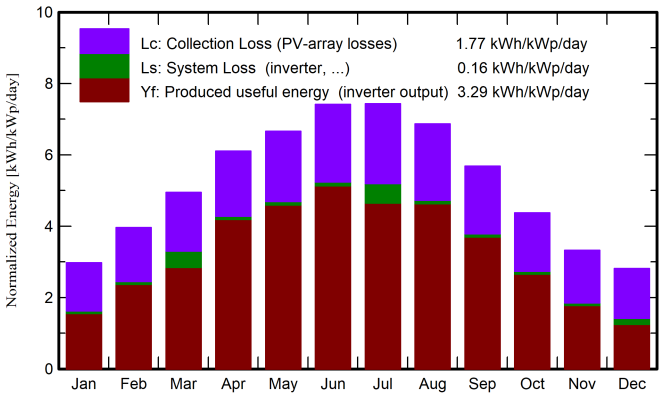
PVsyst V7.4.0
VC3, Simulation date:
26/11/25 23:26
with v7.4.0

Main results

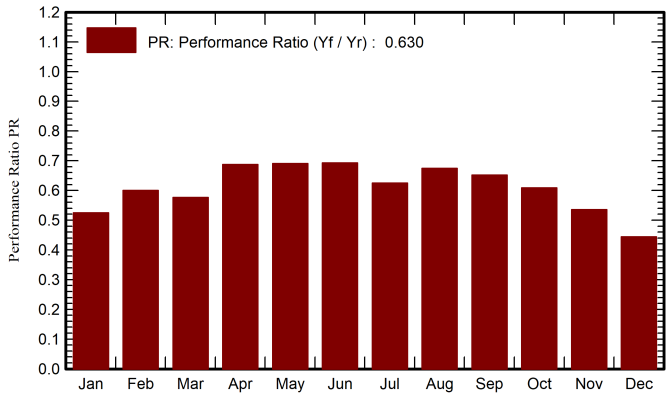
System Production

Produced Energy	1794401 kWh/year	Specific production	1199 kWh/kWp/year
		Perf. Ratio PR	62.97 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	93.2	29.27	9.77	92.1	52.7	75595	72378	0.525
February	111.7	38.35	10.58	110.9	72.4	102851	99484	0.600
March	154.8	55.02	13.32	153.4	110.0	153360	132194	0.576
April	185.1	62.81	15.20	183.1	140.6	192467	188220	0.687
May	208.5	75.23	19.33	206.4	162.8	217875	213306	0.691
June	224.8	70.46	23.12	222.5	178.8	235358	230759	0.693
July	232.9	65.66	26.23	230.3	185.4	240956	215571	0.625
August	214.4	67.66	26.90	212.9	168.1	219440	214952	0.675
September	172.1	54.32	22.78	170.4	127.1	170216	166209	0.652
October	137.0	43.45	19.82	135.6	92.9	127231	123564	0.609
November	100.2	32.55	13.81	99.6	58.8	83149	79779	0.535
December	87.9	24.83	10.94	87.3	46.0	65883	57984	0.444
Year	1922.5	619.63	17.70	1904.3	1395.6	1884382	1794401	0.630

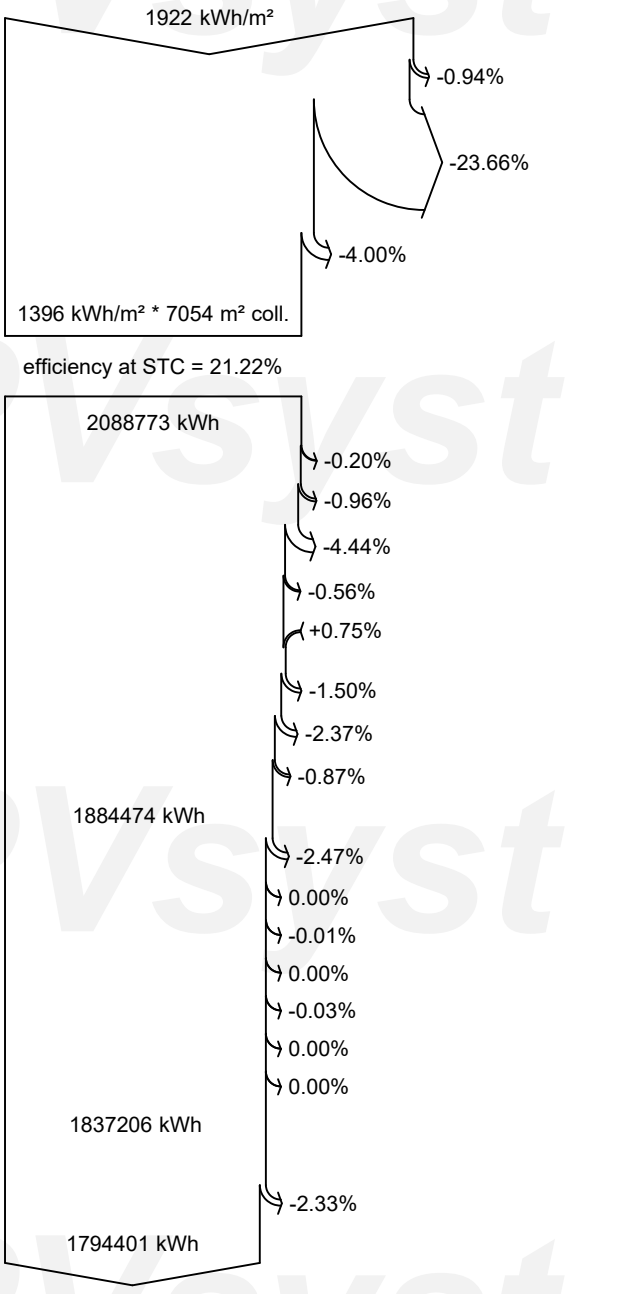
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram



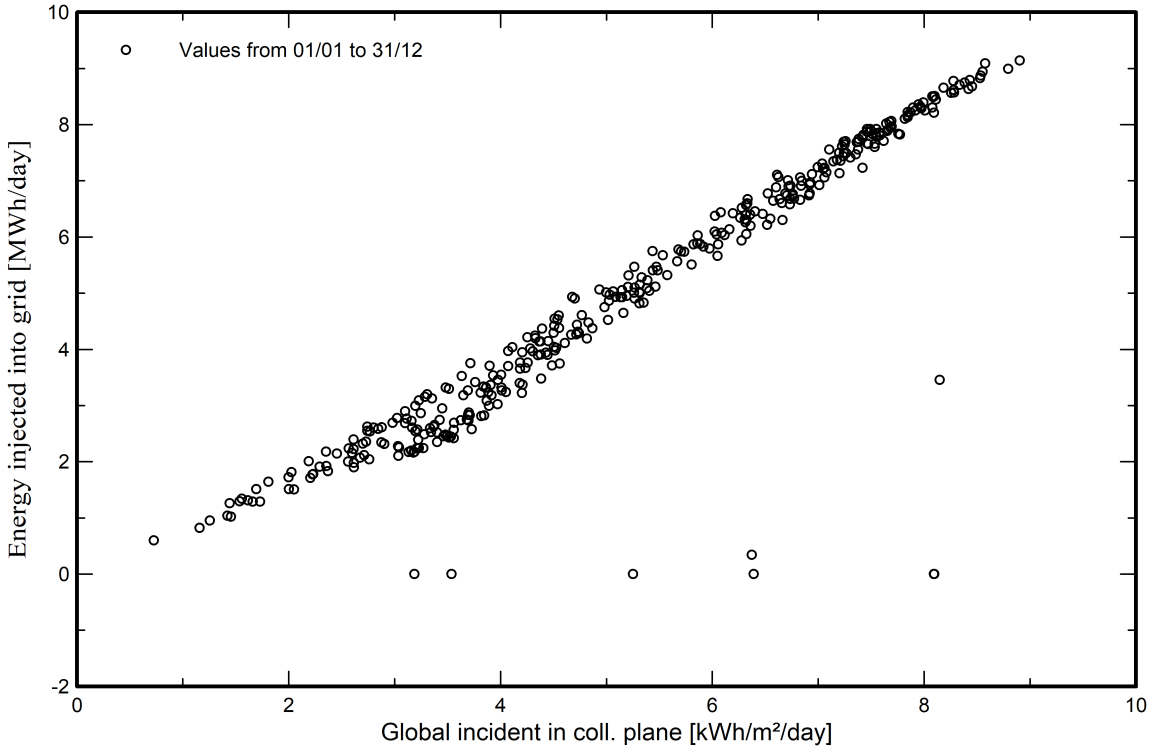
Global horizontal irradiation
Global incident in coll. plane
IAM factor on global
Soiling loss factor
Effective irradiation on collectors
PV conversion
Array nominal energy (at STC effic.)
Module Degradation Loss (for year #1)
PV loss due to irradiance level
PV loss due to temperature
Spectral correction
Module quality loss
LID - Light induced degradation
Mismatch loss, modules and strings
Ohmic wiring loss
Array virtual energy at MPP
Inverter Loss during operation (efficiency)
Inverter Loss over nominal inv. power
Inverter Loss due to max. input current
Inverter Loss over nominal inv. voltage
Inverter Loss due to power threshold
Inverter Loss due to voltage threshold
Night consumption
Available Energy at Inverter Output
System unavailability
Energy injected into grid

PVsyst TRIAL

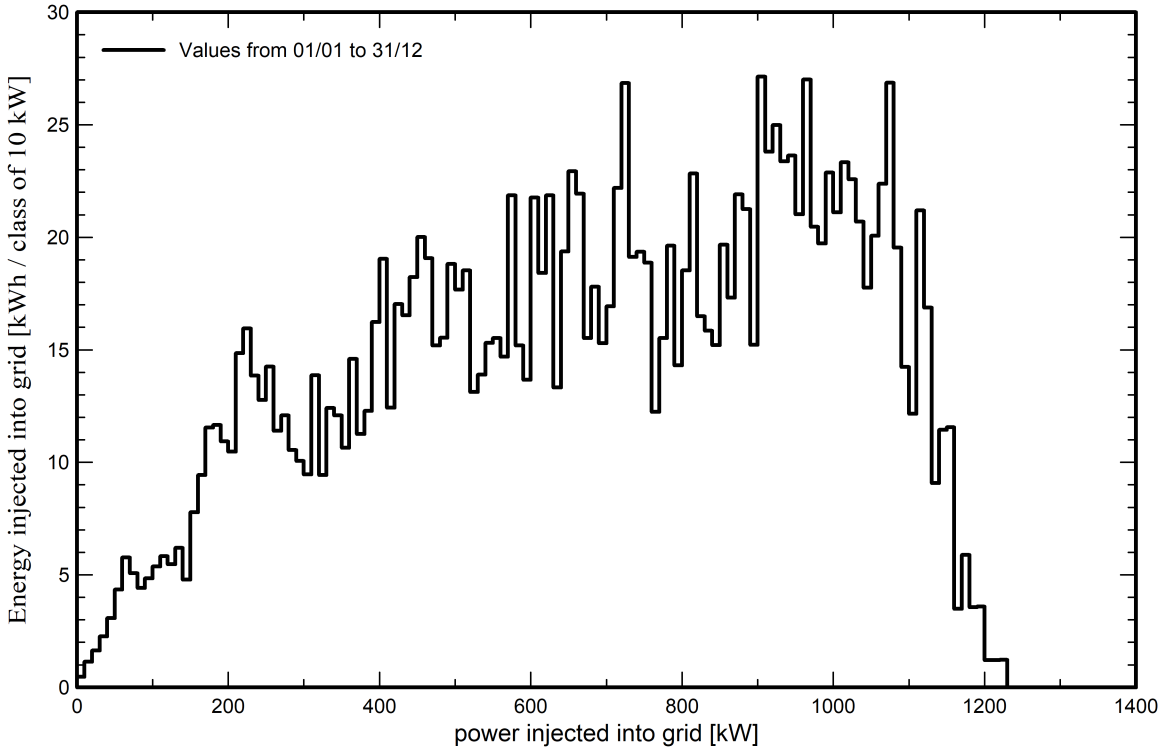
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Predef. graphs

Diagramme d'entrée/sortie journalier



Distribution de la puissance de sortie système



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26/11/25 23:26
with v7.4.0

Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

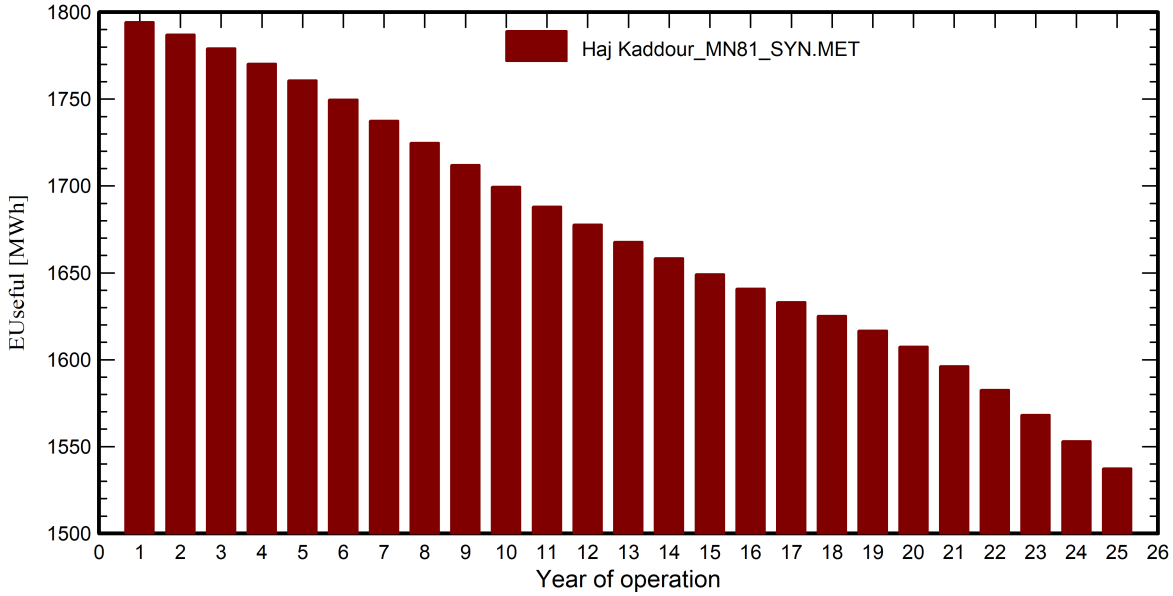
Vmp RMS dispersion 0.4 %/year

Meteo used in the simulation

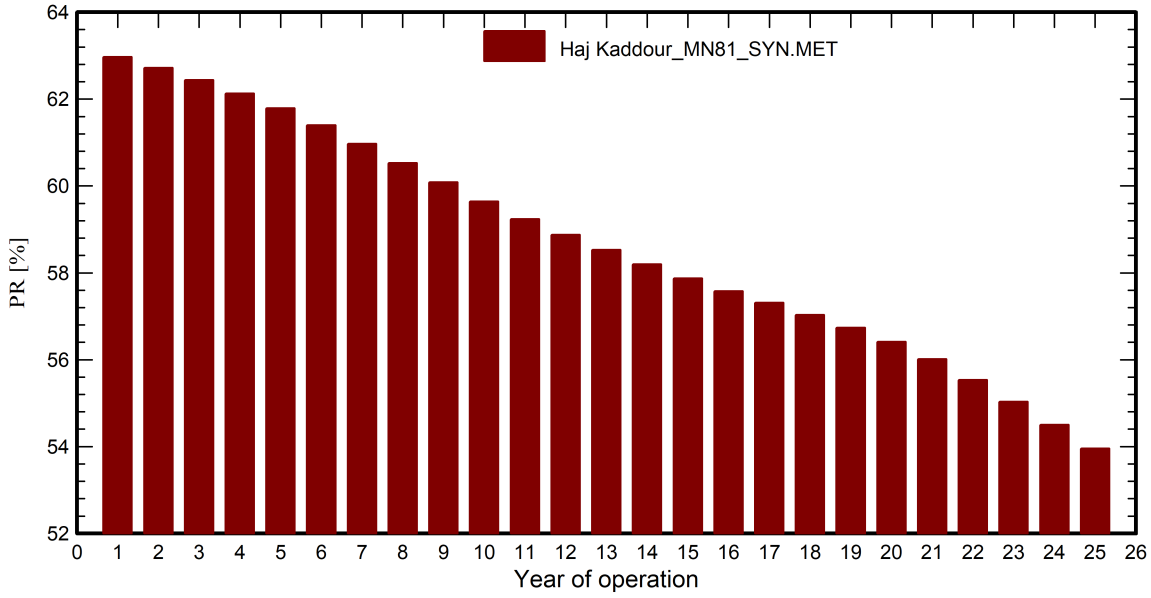
Haj Kaddour MN81 SYN

Years reference year

Useful out system energy



Performance Ratio



PVsyst V7.4.0
VC3, Simulation date:
26/11/25 23:26
with v7.4.0

Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

Meteo used in the simulation

Haj Kaddour MN81 SYN

Years reference year

	EUseful	PR	PR loss
Year	MWh	%	%
1	1794.4	62.97	-0.20
2	1787.2	62.72	-0.60
3	1779.2	62.44	-1.05
4	1770.4	62.13	-1.54
5	1760.7	61.79	-2.07
6	1749.7	61.40	-2.69
7	1737.4	60.97	-3.37
8	1724.8	60.53	-4.07
9	1712.1	60.08	-4.78
10	1699.6	59.64	-5.47
11	1688.1	59.24	-6.11
12	1677.7	58.87	-6.69
13	1667.8	58.53	-7.24
14	1658.3	58.19	-7.77
15	1649.1	57.87	-8.28
16	1640.8	57.58	-8.74
17	1633.1	57.31	-9.17
18	1625.2	57.03	-9.61
19	1616.8	56.74	-10.08
20	1607.5	56.41	-10.59
21	1596.2	56.01	-11.23
22	1582.6	55.54	-11.98
23	1568.1	55.03	-12.78
24	1553.0	54.50	-13.63
25	1537.4	53.95	-14.49

PVsyst TRIAL

PVsyst V7.4.0
VC3, Simulation date:
26/11/25 23:26
with v7.4.0

CO_2 Emission Balance

Total: -2563.1 tCO_2

Generated emissions

Total: 2563.09 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 0.0 tCO_2

System production: 1794.40 MWh/yr

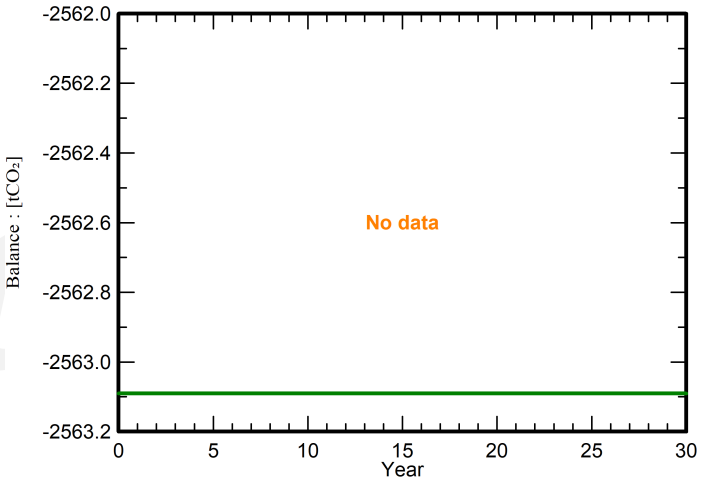
Grid Lifecycle Emissions: 0 gCO_2/kWh

Source: Custom value supplied by user

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	1496 kWp	2562914
Supports	0.01 kgCO2/kg	25800 kg	172
Inverters	0.66 kgCO2/	7.00	4.63

PVsyst TRIAL